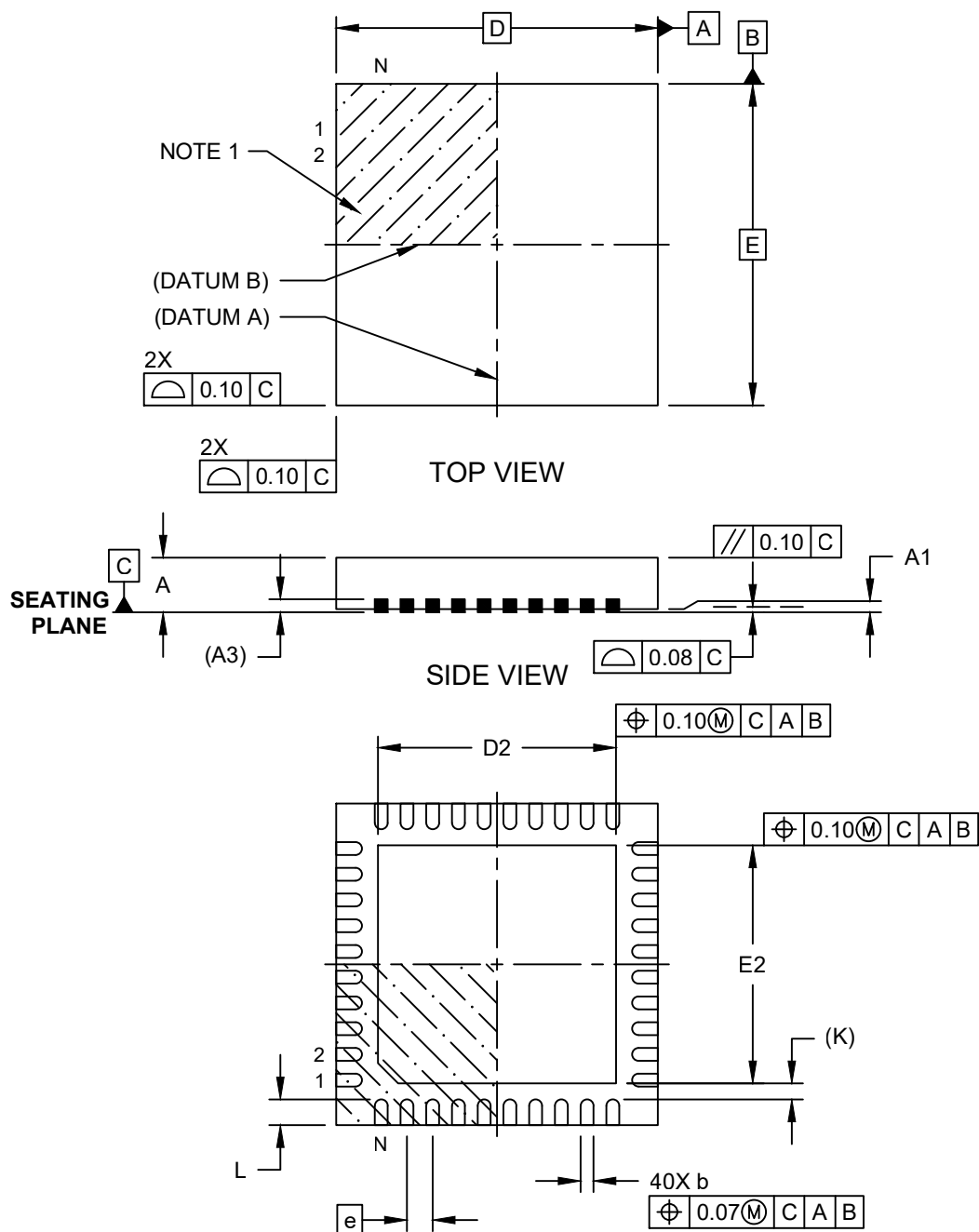


Packaging Diagrams and Parameters

40-Lead Very-Thin Quad Flatpack No-Lead Package (MP) - 5x5x1.0 mm Body [VQFN] With 3.7x3.7 mm Exposed Pad

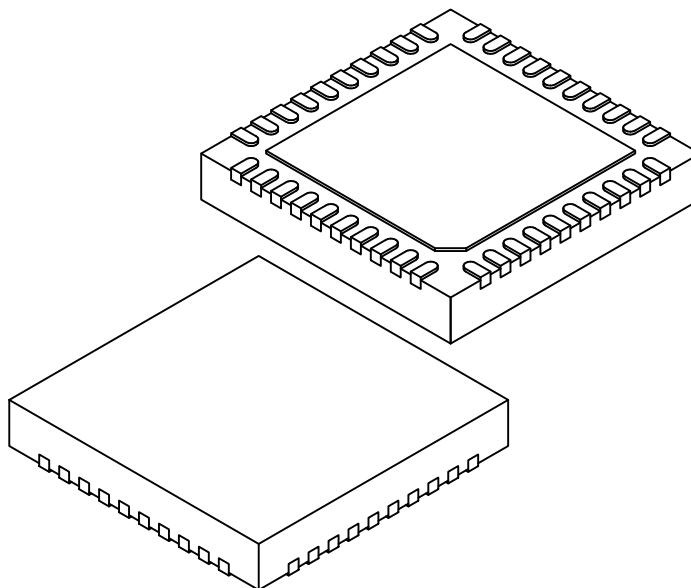
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

40-Lead Very-Thin Quad Flatpack No-Lead Package (MP) - 5x5x1.0 mm Body [VQFN] With 3.7x3.7 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	40		
Pitch	e	0.40 BSC		
Overall Height	A	0.80	0.90	1.00
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3	0.20 REF		
Overall Width	E	5.00 BSC		
Exposed Pad Width	E2	3.60	3.70	3.80
Overall Length	D	5.00 BSC		
Exposed Pad Length	D2	3.60	3.70	3.80
Terminal Width	b	0.15	0.20	0.25
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	0.25 REF		

Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M

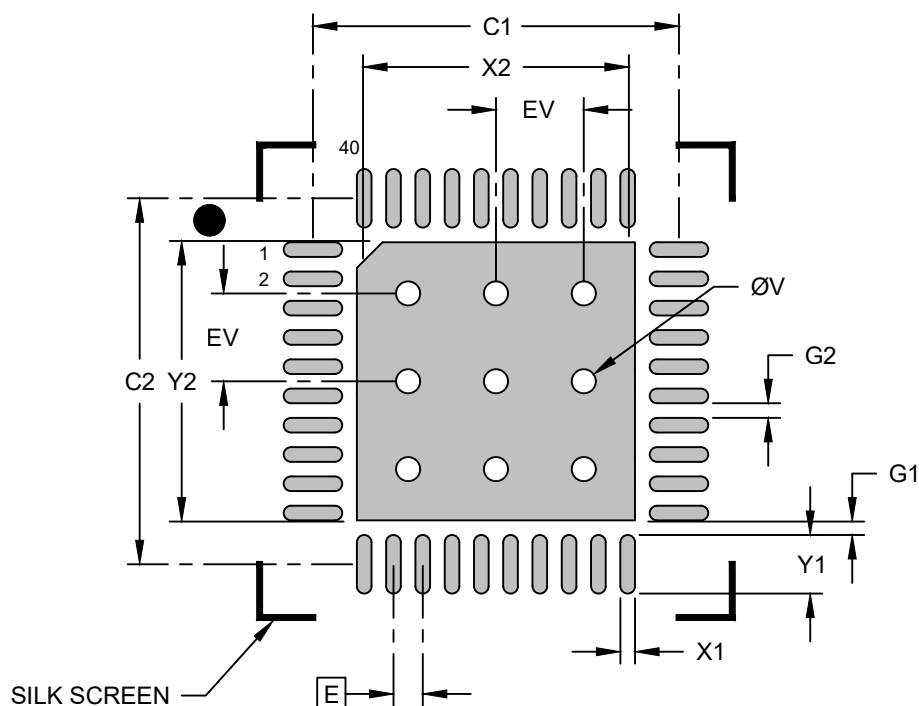
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.

REF: Reference Dimension, usually without tolerance, is for information purposes only.

Land Pattern

40-Lead Very-Thin Quad Flatpack No-Lead Package (MP) - 5x5x1.0 mm Body [VQFN] With 3.7x3.7 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Optional Center Pad Width	X2			3.80
Optional Center Pad Length	Y2			3.80
Contact Pad Spacing	C1		5.00	
Contact Pad Spacing	C2		5.00	
Contact Pad Width (X40)	X1			0.20
Contact Pad Length (X40)	Y1			0.80
Contact Pad to Center Pad (X40)	G1	0.20		
Contact Pad to Contact Pad (X36)	G2	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

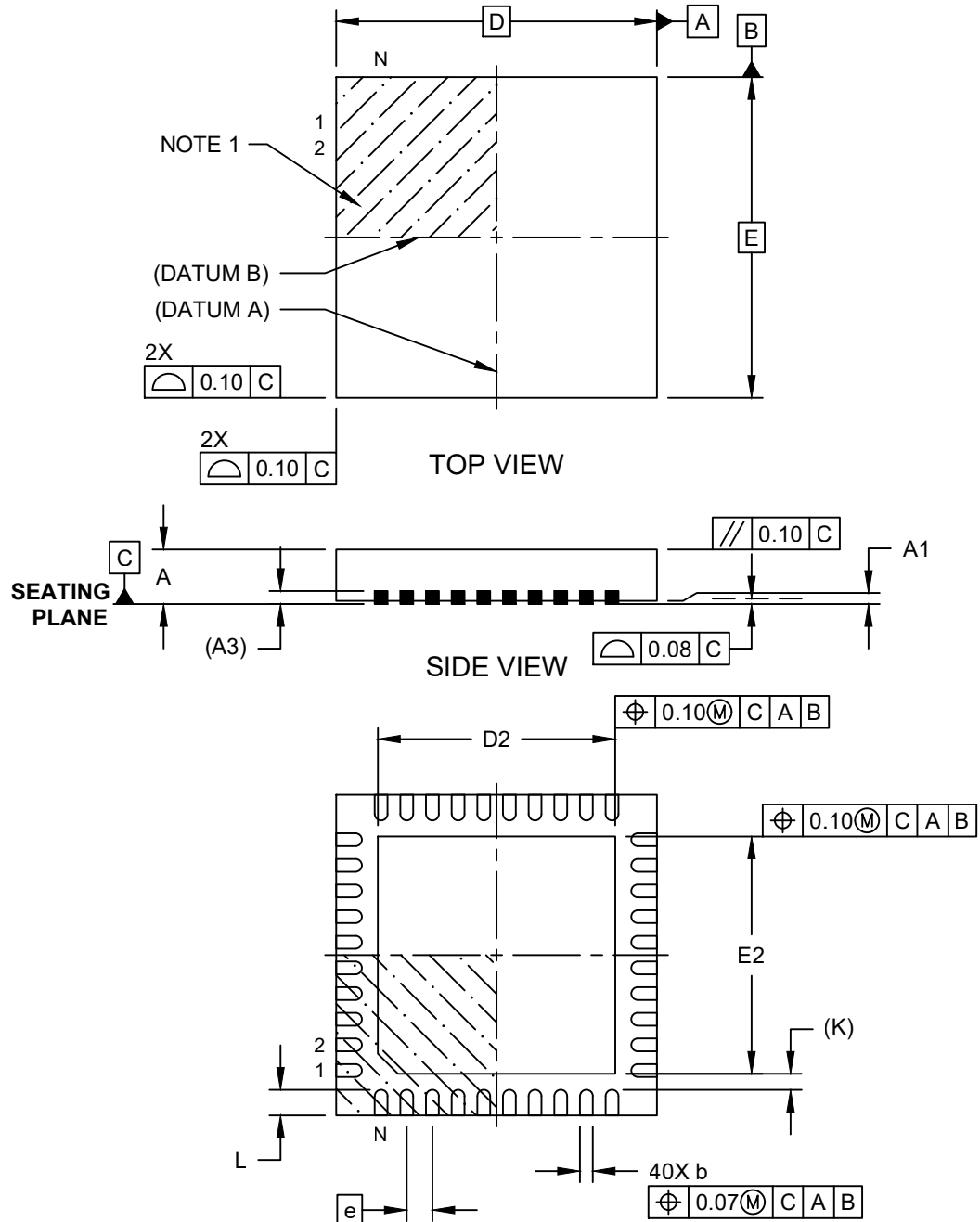
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during the reflow process.

Microchip Technology Drawing C04-02047 (MP) Rev D

Packaging Diagrams and Parameters

40-Lead Very-Thin Quad Flatpack No-Lead Package (Q9X) - 5x5x1.0 mm Body [VQFN] With 3.7x3.7 mm Exposed Pad

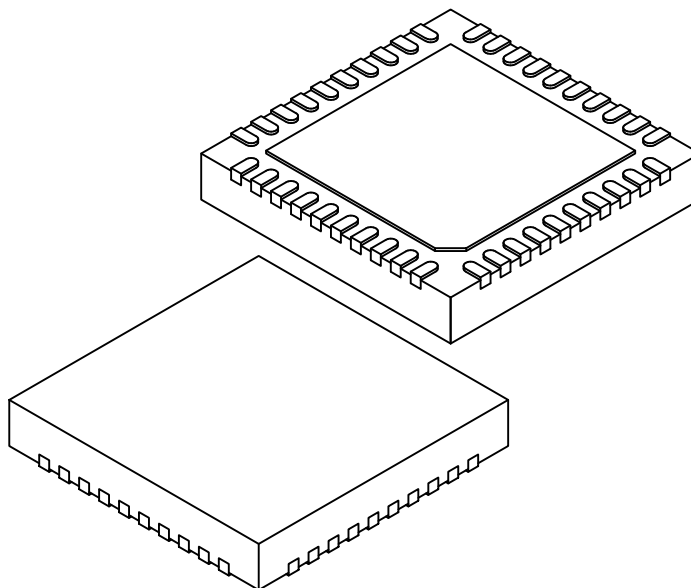
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

40-Lead Very-Thin Quad Flatpack No-Lead Package (Q9X) - 5x5x1.0 mm Body [VQFN] With 3.7x3.7 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	40		
Pitch	e	0.40 BSC		
Overall Height	A	0.80	0.90	1.00
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3	0.20 REF		
Overall Width	E	5.00 BSC		
Exposed Pad Width	E2	3.60	3.70	3.80
Overall Length	D	5.00 BSC		
Exposed Pad Length	D2	3.60	3.70	3.80
Terminal Width	b	0.17	0.20	0.25
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	0.25 REF		

Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M

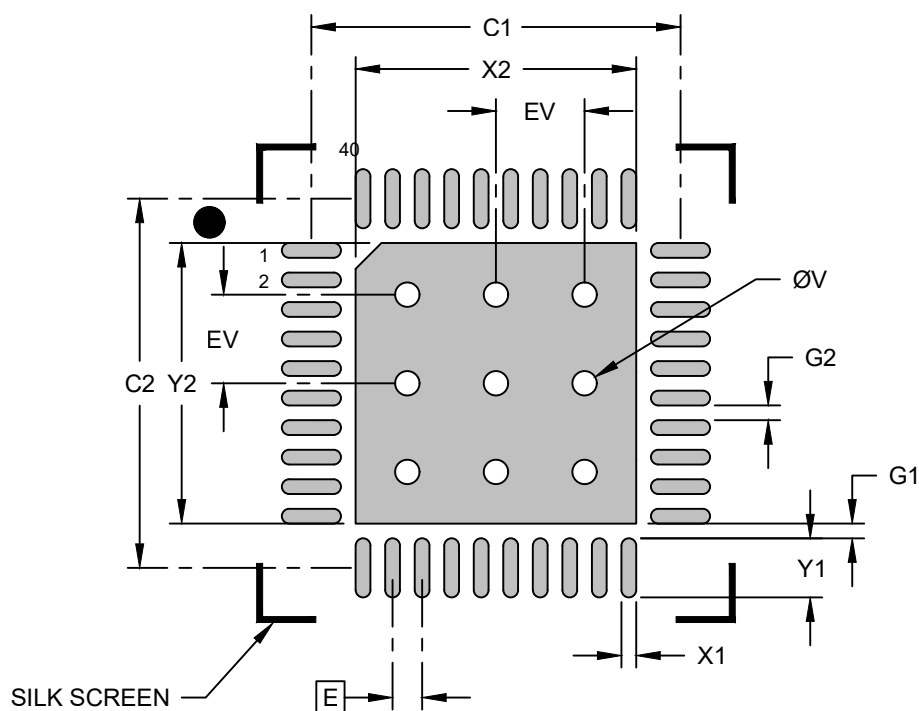
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.

REF: Reference Dimension, usually without tolerance, is for information purposes only.

Land Pattern

40-Lead Very-Thin Quad Flatpack No-Lead Package (Q9X) - 5x5x1.0 mm Body [VQFN] With 3.7x3.7 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.40 BSC		
Optional Center Pad Width	X2			3.80
Optional Center Pad Length	Y2			3.80
Contact Pad Spacing	C1		5.00	
Contact Pad Spacing	C2		5.00	
Contact Pad Width (X40)	X1			0.20
Contact Pad Length (X40)	Y1			0.80
Contact Pad to Center Pad (X40)	G1	0.20		
Contact Pad to Contact Pad (X36)	G2	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during the reflow process.

Microchip Technology Drawing C04-02047 (Q9X) Rev D